P3 dev-v6_aerosols

GEM (HRDPS suite) tests

Description of the tests

In GEM 5.3.0-a15 and MOSA6 configuration

P3 v5.5.0-rc9 vs. P3 v5.5.0 + dev-v6_aerosols

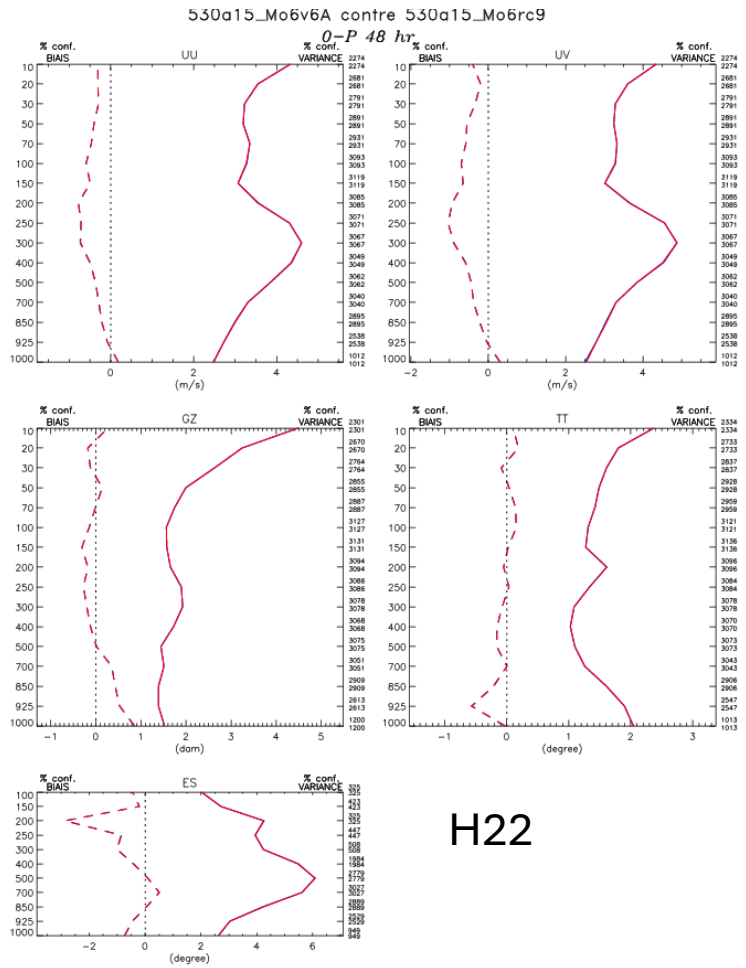
60-61 cases for Winter 2022 and summer 2022

Arcad and emet scores

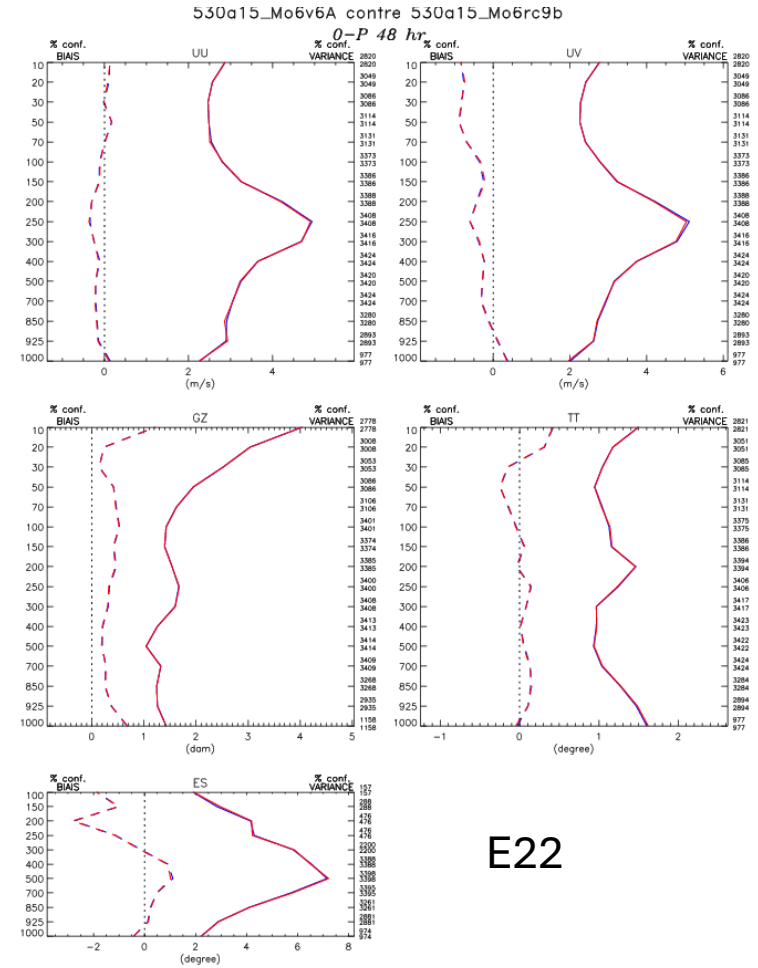
3MOM, liquid fraction,aerosols off.

Note: it will be different because rc9 is not closed 5.5.0.

Arcad



H22

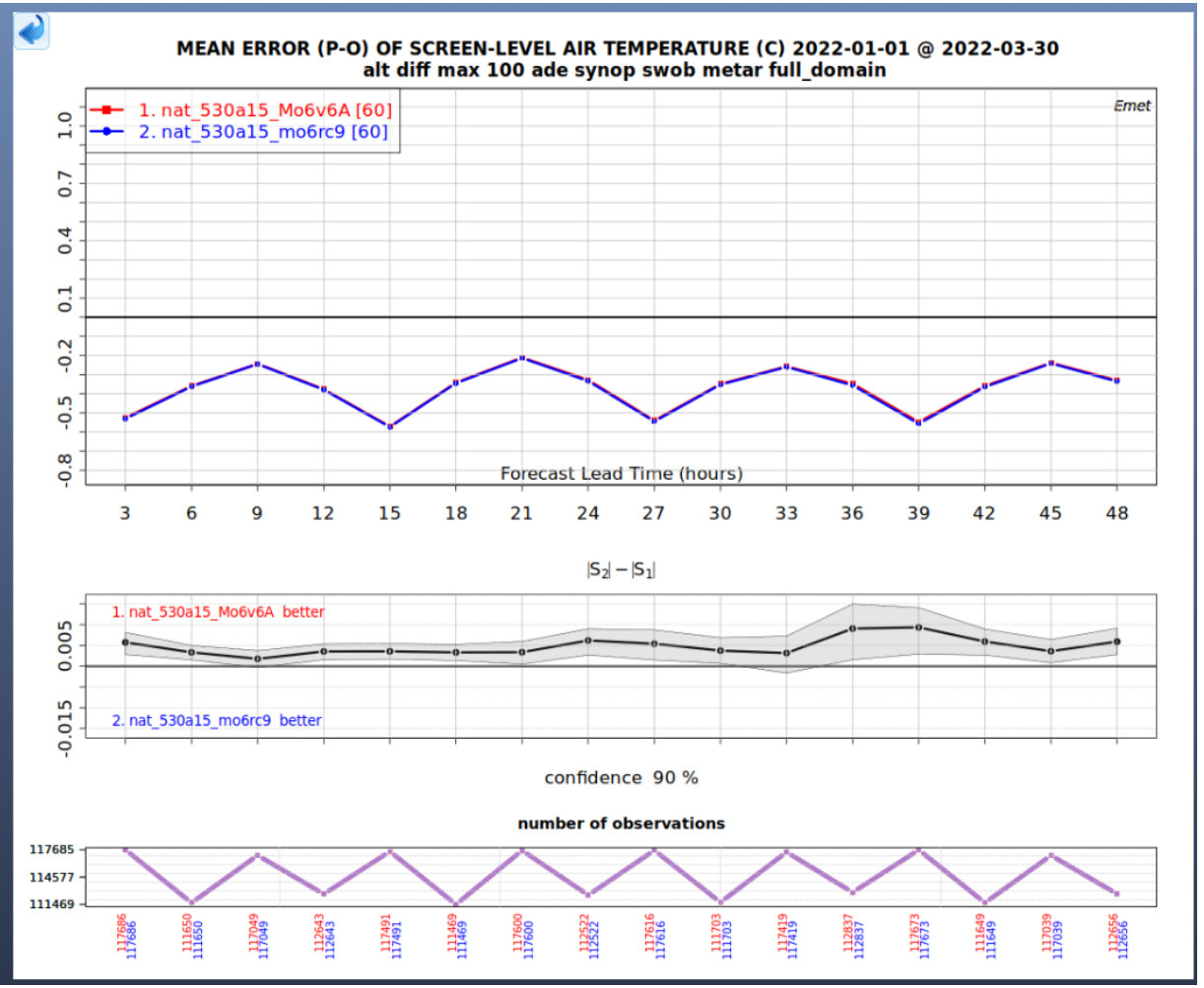


E22

Emet: H22

bias < >

bias nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	-0.0039	-0.0076	-0.0079
	TT	-0.0064	-0.01	-0.0099
Boreal CLIM	TD	-0.00016	-0.0026	-0.00032
	TT	-0.00018	-0.0038	-0.00048
Canada	TD	-6.3e-05	-0.0012	-0.0008
	TT	0.003	0.00024	0.0014
East CLIM	TD	-0.0055	-0.0089	-0.01
	TT	0.0028	0.00036	0.00097
Great Lakes CLIM	TD	-0.0025	-0.0051	-0.005
	TT	0.0028	0.005	0.0025
Mt West CLIM	TD	0.0024	0.0025	0.0032
	TT	0.0012	0.0012	0.0039
North Plains CLIM	TD	0.0012	-0.0016	-0.0004
	TT	0.0014	-0.0015	-0.001
Pacific North West CLIM	TD	-0.016	-0.015	-0.017
	TT	0.015	0.014	0.0094
Prairie CLIM	TD	-0.00028	-5.3e-05	-0.00038
	TT	0.0038	0.0039	0.0028
United States of America East	TD	-0.0033	-0.002	-0.0069
	TT	0.0033	0.0037	0.0022
United States of America West	TD	0.0038	0.00048	0.00069
	TT	0.011	0.0082	0.0029
full_domain	TD	-0.00031	-0.00017	-0.0018
	TT	0.005	0.0059	0.0045



stdev < >

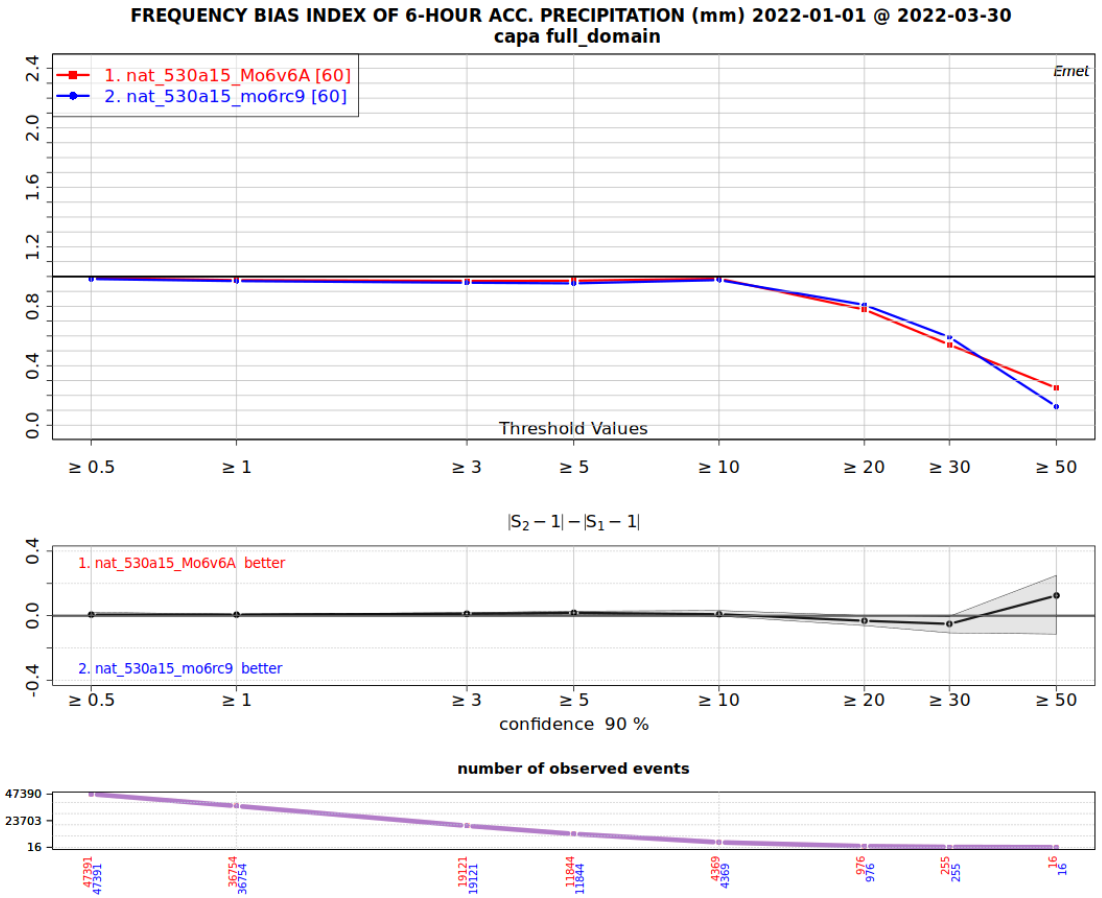
stdev nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	0.0	0.0	0.00087
	TT	0.0	-0.00095	0.0
Boreal CLIM	TD	0.00086	0.0	0.00031
	TT	0.0	0.0	0.0
Canada	TD	0.0018	0.0	0.00075
	TT	0.0	0.0	0.0
East CLIM	TD	0.00084	0.0026	0.0017
	TT	0.0	0.0	0.0
Great Lakes CLIM	TD	0.0011	0.0013	0.00079
	TT	0.0	0.0	0.0
Mt West CLIM	TD	0.0033	0.0019	0.0035
	TT	0.00065	0.00098	0.0022
North Plains CLIM	TD	0.00042	0.0013	0.00095
	TT	0.0	0.001	0.00065
Pacific North West CLIM	TD	0.0029	0.0049	0.0045
	TT	-0.002	-0.0004	-0.00069
Prairie CLIM	TD	0.0016	0.0031	0.004
	TT	0.0	0.0	0.0
United States of America East	TD	0.0014	0.0013	0.0025
	TT	0.0	0.0	0.0
United States of America West	TD	0.0017	0.0	0.0021
	TT	0.0	0.0	0.0
full_domain	TD	0.0013	0.00083	0.0021
	TT	0.0	0.0	0.00023

rmse < >

rmse nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	0.0	0.0	0.0
	TT	0.0	-0.0022	0.0
Boreal CLIM	TD	0.00087	0.0	0.00031
	TT	0.0	0.0	0.0
Canada	TD	0.0017	0.0	0.00056
	TT	0.00058	0.0	0.0
East CLIM	TD	-4.9e-05	0.0	0.0005
	TT	0.0	0.0	0.00042
Great Lakes CLIM	TD	0.0	0.0	0.0
	TT	0.00088	-0.00013	0.0
Mt West CLIM	TD	0.0036	0.0026	0.0036
	TT	0.0017	0.0026	0.0029
North Plains CLIM	TD	0.00091	0.00085	0.00088
	TT	0.0	0.00098	0.00061
Pacific North West CLIM	TD	-0.0013	-0.0012	-0.00032
	TT	-0.0011	-0.00013	0.0001
Prairie CLIM	TD	0.0011	0.0026	0.004
	TT	0.00011	0.0	0.0
United States of America East	TD	0.00045	0.00062	0.00051
	TT	0.00042	0.0012	0.0
United States of America West	TD	0.002	0.0	0.0016
	TT	0.0025	0.0	0.0017
full_domain	TD	0.00094	0.00086	0.0013
	TT	0.00056	0.00063	0.0005

P0 & UV are neutral

Emet: H22



fbi

fbi nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.035
	PR6	0.0	0.0	0.0
Boreal CLIM	PR24	0.015	-0.0035	0.012
	PR6	0.029	0.00073	0.005
Canada	PR24	0.015	-0.0036	-0.0052
	PR6	0.0	-0.0041	-0.0066
East CLIM	PR24	0.0053	-0.0049	0.0
	PR6	0.0023	0.002	0.003
Great Lakes CLIM	PR24	-0.0024	-0.0045	0.0
	PR6	-0.017	-0.00088	-0.0058
Mt West CLIM	PR24	0.0082	0.0	0.0
	PR6	0.0	0.0025	-0.0039
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Pacific North West CLIM	PR24	0.0056	0.0052	0.00027
	PR6	-0.0097	0.0016	-0.0072
Prairie CLIM	PR24	0.0	-0.0023	-0.0066
	PR6	-0.0096	-0.0034	-0.011
United States of America East	PR24	0.005	-0.0072	0.0
	PR6	0.0	0.0012	0.00094
United States of America West	PR24	0.0031	0.0065	0.0052
	PR6	0.01	0.033	0.025
full_domain	PR24	0.0045	0.0027	0.0018
	PR6	0.0047	-0.0025	-0.00098

ets

ets nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	-0.0024	-0.0012
Boreal CLIM	PR24	0.0	-0.003	-0.0028
	PR6	0.016	0.0	0.0067
Canada	PR24	0.0	0.00059	0.0029
	PR6	-0.0006	0.0	-0.00086
East CLIM	PR24	0.00095	-0.0074	-0.0023
	PR6	0.00073	0.0	0.0
Great Lakes CLIM	PR24	-0.0037	0.01	0.0
	PR6	-0.0047	0.0	0.0
Mt West CLIM	PR24	0.0028	0.0	0.0028
	PR6	0.0022	0.0	0.002
North Plains CLIM	PR24	0.0	-0.0034	-0.0026
	PR6	0.0034	0.0	0.0028
Pacific North West CLIM	PR24	-0.0026	0.0022	0.0
	PR6	-0.0039	4.3e-05	-0.0017
Prairie CLIM	PR24	-0.0043	-0.00062	-0.001
	PR6	-0.0023	0.0022	0.0028
United States of America East	PR24	-0.0015	0.0	-0.0018
	PR6	0.0	0.00092	0.0
United States of America West	PR24	0.0	0.0	0.0
	PR6	0.0016	0.0046	0.0022
full_domain	PR24	-0.0025	0.0	-0.00064
	PR6	0.00046	0.0	0.0

pod

pod nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	-0.0018	0.0
Boreal CLIM	PR24	0.0042	0.0025	0.0048
	PR6	0.02	0.0	0.007
Canada	PR24	0.0082	0.0	0.0028
	PR6	0.0017	0.00091	0.0013
East CLIM	PR24	0.0	-0.0091	-0.0026
	PR6	0.0014	0.00067	0.00089
Great Lakes CLIM	PR24	0.0	0.0034	0.0
	PR6	-0.018	0.0036	0.0
Mt West CLIM	PR24	0.0028	0.0	0.0017
	PR6	0.0	0.0023	0.0019
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0065	0.0	0.0
Pacific North West CLIM	PR24	0.0	0.0081	0.005
	PR6	-0.0041	0.0062	-0.0039
Prairie CLIM	PR24	-0.0051	0.0	0.0
	PR6	-0.0031	0.00099	-0.0028
United States of America East	PR24	0.0	-0.0065	-0.002
	PR6	0.0	0.0019	-0.0036
United States of America West	PR24	0.0	0.0068	0.0048
	PR6	0.0042	-0.013	-0.01
full_domain	PR24	0.0		
	PR6	0.0011		

far

far nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0061	-0.0023	-0.0013
Boreal CLIM	PR24	-0.0073	0.0	-0.002
	PR6	0.016	0.0	0.009
Canada	PR24	-0.002	0.00094	0.0023
	PR6	-0.0027	-0.0028	-0.0036
East CLIM	PR24	-0.0041	-0.016	-0.0041
	PR6	-0.00079	0.0	-0.00046
Great Lakes CLIM	PR24	-0.00048	0.012	0.0
	PR6	-0.0092	-0.0018	-0.0014
Mt West CLIM	PR24	-0.0043	0.0	0.0
	PR6	0.0049	0.0	0.0018
North Plains CLIM	PR24	0.0	-0.0037	-0.0028
	PR6	0.0044	0.0	0.0033
Pacific North West CLIM	PR24	-0.0034	-0.0048	-0.0051
	PR6	-0.005	-0.0056	-0.0061
Prairie CLIM	PR24	-0.0037	0.00043	0.0
	PR6	0.0	0.012	0.0089
United States of America East	PR24	-0.0018	-0.0011	-0.00091
	PR6	0.0	0.0	-0.00079
United States of America West	PR24	-0.002	-0.0052	-0.0054
	PR6	0.0	-0.0014	-0.0018
full_domain	PR24	-0.0032	0.0005	-0.0012
	PR6	-0.002	0.0013	-0.0022

fbi clrfew 0 - 2

fbi_clrfew_0-2 nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TCC	0.0032	0.0	0.0
Boreal CLIM	TCC	0.0	0.00062	-0.0041
Canada	TCC	0.0	-0.0027	-0.0036
East CLIM	TCC	-0.0015	-0.0016	-0.0014
Great Lakes CLIM	TCC	-0.0063	-0.0083	-0.0049
Mt West CLIM	TCC	-0.0035	0.0	0.0
North Plains CLIM	TCC	0.0	0.0	0.0
Pacific North West CLIM	TCC	-0.0031	-0.0056	-0.00052
Prairie CLIM	TCC	0.0	-0.0011	-0.002
United States of America East	TCC	-0.0015	-0.0024	-0.0026
United States of America West	TCC	-0.0031	0.0	-0.0022
full_domain	TCC	-0.0012	-0.0026	-0.0022

far clrfew 0 - 2

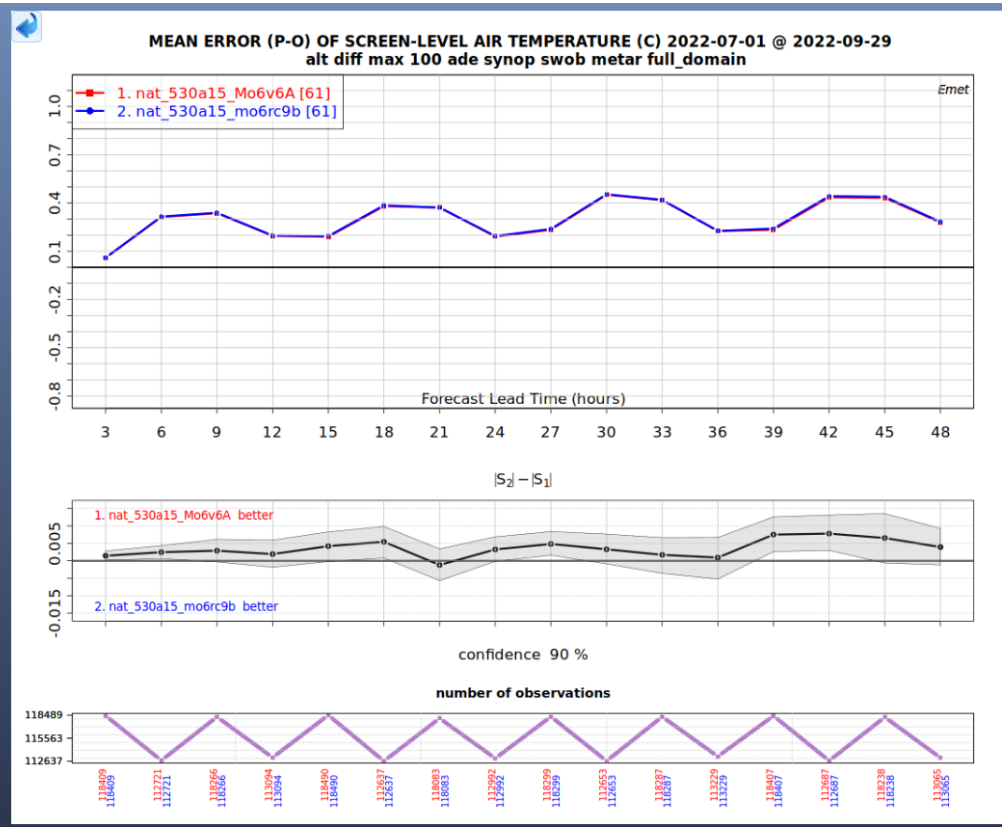
far_clrfew_0-2 nat_530a15_Mo6v6A / nat_530a15_mo6rc9		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TCC	0.0	0.0	0.0
Boreal CLIM	TCC	-0.0011	0.0012	0.00024
Canada	TCC	-0.00097	0.0016	0.00036
East CLIM	TCC	0.0017	0.00063	0.0
Great Lakes CLIM	TCC	0.0049	0.0026	0.0016
Mt West CLIM	TCC	0.0	0.0	-7.9e-06
North Plains CLIM	TCC	0.0	0.0	0.0
Pacific North West CLIM	TCC	0.0005	0.0067	0.0025
Prairie CLIM	TCC	0.0021	0.00017	0.00057
United States of America East	TCC	0.0015	0.0019	0.0016
United States of America West	TCC	0.0042	0.0026	0.005
full_domain	TCC	0.0011	0.00053	0.00062

PR6, PR24, TCC mostly neutral

Emet: E22

bias

bias nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	TD	-0.0025	0.0079	-0.002
	TT	-0.0016	-0.001	0.0016
Boreal CLIM	TD	-0.0021	0.0051	0.0047
	TT	-0.011	-0.00067	-0.0029
Canada	TD	0.0027	0.0035	0.0047
	TT	-0.0083	-0.0031	-0.0036
East CLIM	TD	0.0065	0.0046	0.0084
	TT	0.011	0.0066	0.0086
Great Lakes CLIM	TD	0.0037	0.0	0.001
	TT	-0.0033	0.00047	0.00017
Mt West CLIM	TD	-0.0039	-0.0037	-0.0069
	TT	-0.0012	-0.00024	-0.00035
North Plains CLIM	TD	-0.0016	0.0	-0.00019
	TT	-0.002	-0.0033	-0.0019
Pacific North West CLIM	TD	-0.00019	0.0011	0.00052
	TT	-0.0089	-0.01	-0.0028
Prairie CLIM	TD	0.00029	0.0037	0.0044
	TT	0.016	0.0076	0.011
United States of America East	TD	0.0062	0.0046	0.009
	TT	0.013	0.007	0.0082
United States of America West	TD	0.0017	-0.0033	0.0051
	TT	0.0073	0.0079	0.0059
full_domain	TD	0.0011	0.00099	0.002
	TT	0.0029	0.0025	0.0018



rmse

rmse nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	TD	-0.005	-0.00035	0.0
	TT	-0.0095	-0.0028	-0.0053
Boreal CLIM	TD	-0.0027	-0.0023	-0.0035
	TT	-0.0059	-0.0061	-0.0061
Canada	TD	-0.003	-0.0032	-0.0034
	TT	-0.0087	-0.0038	-0.0075
East CLIM	TD	0.0027	0.0023	0.0017
	TT	-0.0027	-0.0014	-0.0021
Great Lakes CLIM	TD	-0.0018	0.0	0.0
	TT	-0.0042	-0.0017	-0.0039
Mt West CLIM	TD	-0.0053	-0.0073	-0.0067
	TT	-0.0033	0.0012	0.0
North Plains CLIM	TD	-0.0039	0.0	-0.0032
	TT	-0.0043	-0.00079	-0.0029
Pacific North West CLIM	TD	0.0	0.0	0.00054
	TT	0.0007	0.0016	0.00047
Prairie CLIM	TD	0.00099	0.0	0.00062
	TT	0.0	0.002	0.0006
United States of America East	TD	0.0017	0.0017	0.001
	TT	0.0	1.1e-05	-0.00058
United States of America West	TD	-0.0025	-0.0063	-0.0053
	TT	-0.0021	0.0	0.0
full_domain	TD	-0.0014	-0.00039	-0.0012
	TT	-0.0022	-0.0016	-0.0022

stdev

stdev nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	TD	-0.0044	0.0	0.0
	TT	-0.008	-0.0025	-0.0055
Boreal CLIM	TD	-0.0055	-0.0018	-0.0042
	TT	-0.0021	-0.0038	-0.0037
Canada	TD	-0.0027	-0.0031	-0.0036
	TT	-0.0053	-0.0015	-0.0063
East CLIM	TD	0.0011	0.0	0.00029
	TT	-0.006	-0.0025	-0.0026
Great Lakes CLIM	TD	0.0	0.0	0.0
	TT	-0.0041	0.0	-0.0031
Mt West CLIM	TD	-0.0057	-0.0069	-0.0055
	TT	-0.004	0.00067	0.0
North Plains CLIM	TD	-0.0041	0.0	-0.0032
	TT	0.0	0.0	-0.00093
Pacific North West CLIM	TD	0.0	0.0	0.0
	TT	0.00053	0.002	0.00031
Prairie CLIM	TD	0.001	0.0	0.0013
	TT	0.00067	0.0011	0.0
United States of America East	TD	0.00063	0.0	0.0
	TT	-0.0026	-0.00017	-0.0021
United States of America West	TD	-0.0038	-0.0064	-0.0055
	TT	-0.0031	0.0	0.0
full_domain	TD	-0.0013	-0.00049	-0.0013
	TT			-0.0027

bias

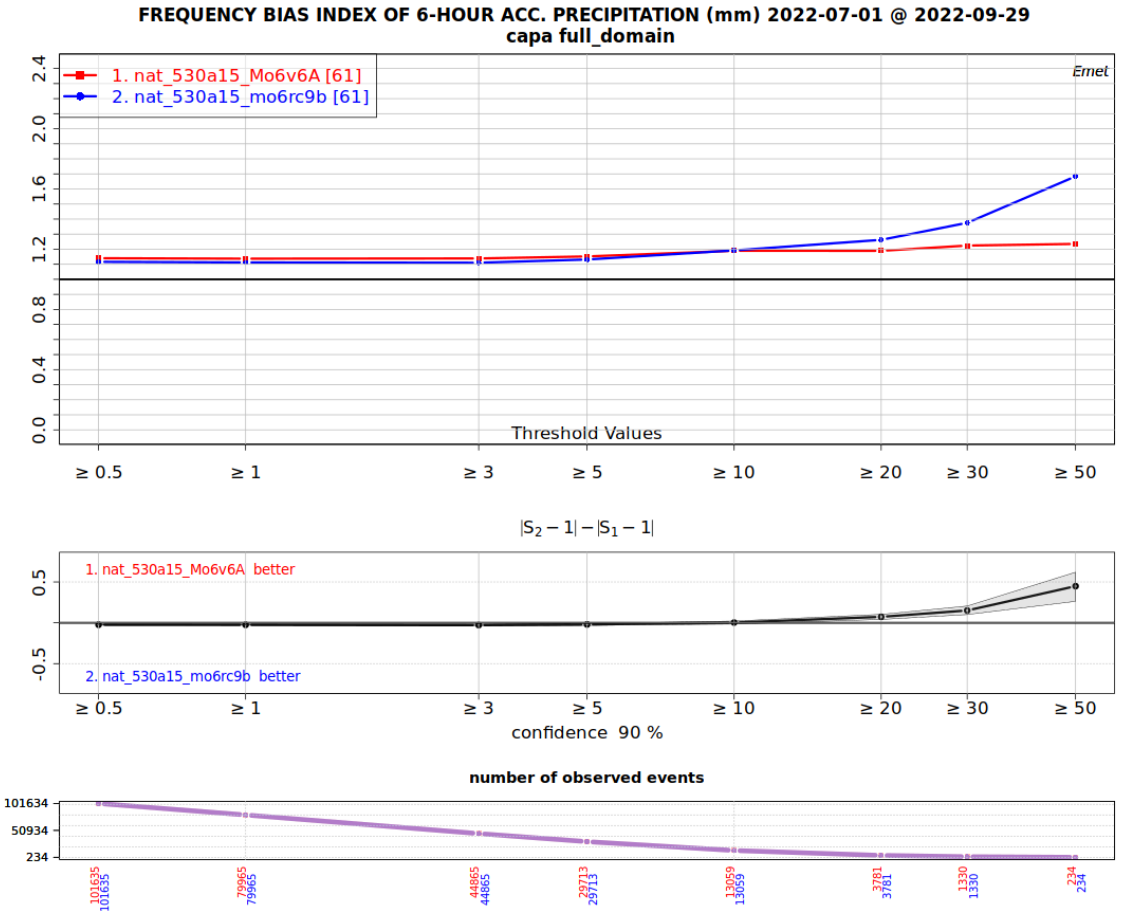
bias nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	P0	0.0044	0.00082	0.0027
Boreal CLIM	P0	0.00082	-0.0027	-0.00062
Canada	P0	-0.00035	-0.002	-0.00022
East CLIM	P0	0.00098	-0.0029	-0.0049
Great Lakes CLIM	P0	0.0036	0.014	0.0076
Mt West CLIM	P0	0.0044	-6.5e-06	0.00057
North Plains CLIM	P0	-0.00092	0.0022	-0.00077
Pacific North West CLIM	P0	-0.0061	-0.0068	-0.011
Prairie CLIM	P0	0.0	-0.00092	0.0
United States of America East	P0	0.0067	0.011	0.01
United States of America West	P0	-0.0016	-0.0016	-0.002
full_domain	P0	-0.00097	-0.0018	-0.0011

bias

bias nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	UV	0.003	-0.00087	0.001
Boreal CLIM	UV	0.0	0.0036	0.0025
Canada	UV	0.00098	0.0016	-0.00039
East CLIM	UV	0.0041	0.00062	0.0041
Great Lakes CLIM	UV	0.0	0.00092	-0.00032
Mt West CLIM	UV	-0.0063	0.0011	-0.0053
North Plains CLIM	UV	0.0025	0.0016	0.0017
Pacific North West CLIM	UV	0.0	-0.0016	-0.0013
Prairie CLIM	UV	0.00032	0.0014	0.0
United States of America East	UV	0.0021	0.0032	0.0027
United States of America West	UV	0.0026	0.0	-0.0015
full_domain	UV	0.0022	0.0024	0.0019

P0 & UV are neutral

Emet: E22



fbi

fbi nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	-0.015
	PR6	-0.013	-0.057	-0.057
Boreal CLIM	PR24	0.0	-0.0096	-0.0014
	PR6	-0.057	-0.021	-0.008
Canada	PR24	0.0	-0.0054	-0.004
	PR6	-0.0013	0.003	0.009
East CLIM	PR24	0.01	0.056	0.04
	PR6	0.059	0.056	0.068
Great Lakes CLIM	PR24	0.013	0.034	0.008
	PR6	-0.0092	0.0	-0.0067
Mt West CLIM	PR24	0.0	0.037	-0.0028
	PR6	-0.0031	-0.019	0.073
North Plains CLIM	PR24	0.019	-0.0058	0.018
	PR6	0.13	0.045	0.0076
Pacific North West CLIM	PR24	0.0	-0.0075	-0.013
	PR6	-0.017	-0.012	-0.13
Prairie CLIM	PR24	0.0	0.019	-0.002
	PR6	-0.004	0.13	0.11
United States of America East	PR24	0.0075	0.047	0.045
	PR6	0.076	0.099	0.089
United States of America West	PR24	0.0	0.044	0.028
	PR6	0.039	0.095	0.052
full_domain	PR24	0.0079	0.037	0.029
	PR6	0.087	0.059	0.072

ets

ets nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0016	0.0	0.0011
Boreal CLIM	PR24	-0.0023	-0.0045	-0.0017
	PR6	-0.0019	-0.001	-0.0026
Canada	PR24	-0.00095	-0.0048	0.0
	PR6	0.0011	-0.007	-0.0014
East CLIM	PR24	0.0035	0.0	0.0021
	PR6	0.0045	0.0	0.0029
Great Lakes CLIM	PR24	0.0	0.0	0.0
	PR6	0.01	0.0	0.0044
Mt West CLIM	PR24	0.0	-0.00092	0.0
	PR6	0.0	0.0	0.0017
North Plains CLIM	PR24	-0.014	0.0	-0.013
	PR6	0.0028	0.0044	0.00067
Pacific North West CLIM	PR24	0.019	-0.0015	0.0074
	PR6	0.003	0.0	-0.005
Prairie CLIM	PR24	0.0023	0.0	0.0
	PR6	0.0	0.004	0.0024
United States of America East	PR24	0.0	0.0	0.0
	PR6	0.0082	0.0021	0.0052
United States of America West	PR24	-0.0032	0.0	0.0
	PR6	0.0	0.0	0.0
full_domain	PR24	0.002	0.0	0.0
	PR6	0.0018	0.00049	0.0012

far

far nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Boreal CLIM	PR24	-0.0015	-0.011	-0.0017
	PR6	-0.0013	-0.0026	-0.0018
Canada	PR24	0.0	-0.013	-0.0066
	PR6	0.0	-0.015	-0.0047
East CLIM	PR24	0.0039	0.0055	0.0044
	PR6	0.0034	0.0	0.0028
Great Lakes CLIM	PR24	0.0	0.0	-0.00084
	PR6	0.013	-0.0013	0.0036
Mt West CLIM	PR24	0.0	-0.0013	-0.00059
	PR6	0.0	-0.0018	0.0023
North Plains CLIM	PR24	-0.019	-0.0018	-0.018
	PR6	0.0	0.002	-0.00081
Pacific North West CLIM	PR24	-0.0056	-0.0027	-0.0043
	PR6	-0.0057	-0.0019	-0.013
Prairie CLIM	PR24	-0.0015	0.0	0.0
	PR6	0.0	0.0056	0.0073
United States of	PR24	0.0	0.0034	0.0026

pod

pod nat_530a15_Mo6v6A / nat_530a15_mo6rc9b		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0049	0.0	0.0021
Boreal CLIM	PR24	0.0	0.0038	0.00081
	PR6	-0.004	0.0027	-0.0047
Canada	PR24	0.00066	0.0	0.00051
	PR6	0.0016	-0.0063	-0.0011
East CLIM	PR24	0.0	0.0018	0.0018
	PR6	0.0091	0.0051	0.0072
Great Lakes CLIM	PR24	0.0063	-0.0059	0.0
	PR6	0.013	0.0	0.002
Mt West CLIM	PR24	0.0	0.0	0.0014
	PR6	0.0	0.003	0.0057
North Plains CLIM	PR24	-0.025	0.0015	-0.018
	PR6	0.0014	0.0098	0.0013
Pacific North West CLIM	PR24	0.031	0.009	0.015
	PR6	0.014	0.0025	0.0099
Prairie CLIM	PR24	0.0	0.0	0.0015
	PR6	0.0	0.0	-0.0032
United States of America East	PR24	0.0012	0.002	0.0026
	PR6	0.0081	0.0057	0.0079
United States of America West	PR24	-0.01	0.0	0.0
	PR6	0.0	0.0	0.0015
full_domain	PR24	0.00084	-0.00089	-0.00083
	PR6	0.0054	0.00052	0.005

I believe this is due to the change of autoconversion parameterization coming from RPN-A: Checking link

Conclusions

- Completely neutral compared to v5.5.0-rc9 when aerosols are off.
- Reduction of large PR thresholds for PR6/PR24 FBI during summer: consequence of RPN-A changes and not the aerosols off code.